1. Problem Statement

Develop a C++ program using DFS (Depth First Search) to identify disconnected components in a computer network.

2. Objectives

1. To represent a computer network using a graph.
2. To implement the DFS (Depth First Search) algorithm in C++.
3. To identify connected components in a computer network.
4. To detect isolated computers and disconnected groups.
5. To count and display the total number of connected components.
6. To understand the application of graph traversal in network connectivity analysis.

3. Introduction

A computer network consists of computers connected to each other to exchange information and share resources. Sometimes, all computers in a network are not directly or indirectly connected, resulting in separate groups.

Graph theory is used to represent computer networks. In a graph, computers are represented as vertices, and connections between computers are represented as edges.

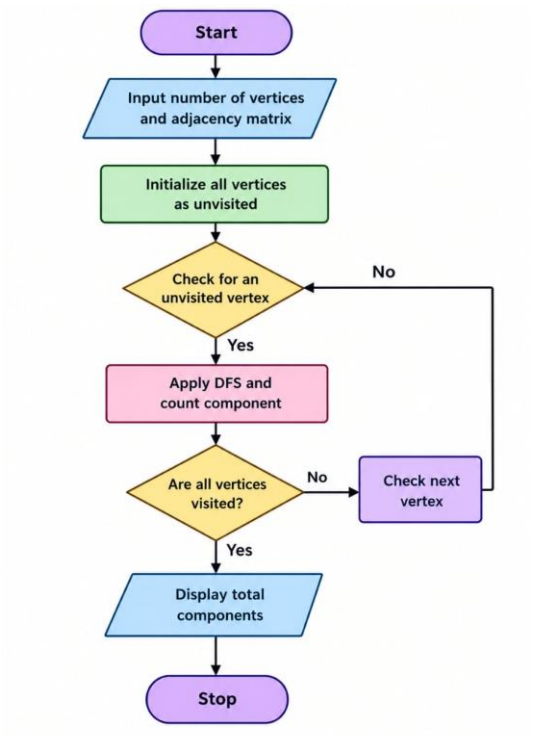
Depth First Search (DFS) is a graph traversal algorithm that explores a vertex and its unvisited neighboring vertices before returning to previously visited vertices.

A connected component is a group of vertices in which every vertex is reachable from the other vertices in that group. Using DFS, we can identify separate connected components and count them. This helps us understand the connectivity of a computer network.

4. Algorithm

1. Start.
2. Declare an adjacency matrix to represent the graph.
3. Input the number of vertices and the adjacency matrix.
4. Initialize all vertices as unvisited.
5. Traverse each vertex of the graph.
6. If a vertex is unvisited, increment the component count.
7. Apply DFS from that vertex and mark all reachable vertices as visited.
8. Display the vertices belonging to that component.
9. Repeat until all vertices are checked.
10. Display the total number of connected components.
11. Stop.

5. Flowchart



9.Result

The C++ program was successfully developed using the DFS algorithm to identify connected components in a computer network. For the given sample input, the program identifies three connected components and displays their vertices.

10.Conclusion

The experiment demonstrates the use of DFS for identifying connected components in a computer network. By traversing all vertices and checking their connections, the program determines the number of separate groups. This technique is useful for analyzing network connectivity and identifying isolated computers.